COLORADO STATE UNIVERSITY FORECAST OF ATLANTIC HURRICANE ACTIVITY FROM OCTOBER 14–27, 2026

We are favoring below-normal Atlantic hurricane activity for the next two weeks (60%), with a lower chance of near-normal (38%) activity and a very low chance of above-normal activity (2%) in the next two weeks.

(as of 14 October 2026)

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In Memory of William M. Gray[8]

This discussion as well as past forecasts and verifications are available online at http://tropical.colostate.edu

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1 Introduction

This is the 18th year that we have issued shorter-term forecasts of tropical cyclone (TC) activity starting in early August. These two-week forecasts are based on a combination of observational and modeling tools. The primary tools that are used for this forecast are as follows: 1) current storm activity, 2) National Hurricane Center Tropical Weather Outlooks, 3) forecast output from global models, 4) the current and projected state of the Madden-Julian oscillation (MJO) and 5) the current seasonal forecast.

Our forecast definition of above-normal, normal, and below-normal Accumulated Cyclone Energy (ACE) periods is defined by ranking observed activity in the satellite era from 1966–2025 and defining above-normal, normal and below-normal two-week periods based on terciles. Since there are 60 years from 1966–2025, we include 20 years in each tercile.

Table 1: ACE forecast definition and probabilistic forecast for TC activity for October 14–27, 2026.

Parameter	Definition	Probability in Each Category
Above-Normal	Upper Tercile (>8 ACE)	2%
Normal	Middle Tercile (1–8 ACE)	38%
Below-Normal	Lower Tercile (<1 ACE)	60%

2 Forecast

We are favoring below-normal activity (<1 ACE) during the next two weeks, although relatively little TC activity is required to reach the normal category (1–7 ACE). The National Hurricane Center is currently not monitoring any areas of TC formation in the next week. There is a weak signal for a system forming in the subtropical Atlantic during week two of our forecast period. The MJO is forecast to propagate across Africa and the Indian Ocean over the next two weeks. When the MJO is in these phases, it tends to enhance Atlantic TC activity. However, the base state across the Atlantic is quite TC-unfavorable, given the strong El Niño and associated high levels of vertical wind shear.

Figure 1 displays the formation locations of TCs from October 14–27 for the years from 1966–2025, along with the maximum intensities that these storms reached. The primary threat area for major hurricane formations during mid- to late October is in the western Caribbean. Figure 2 displays the October 14–27 forecast period with respect to climatology.

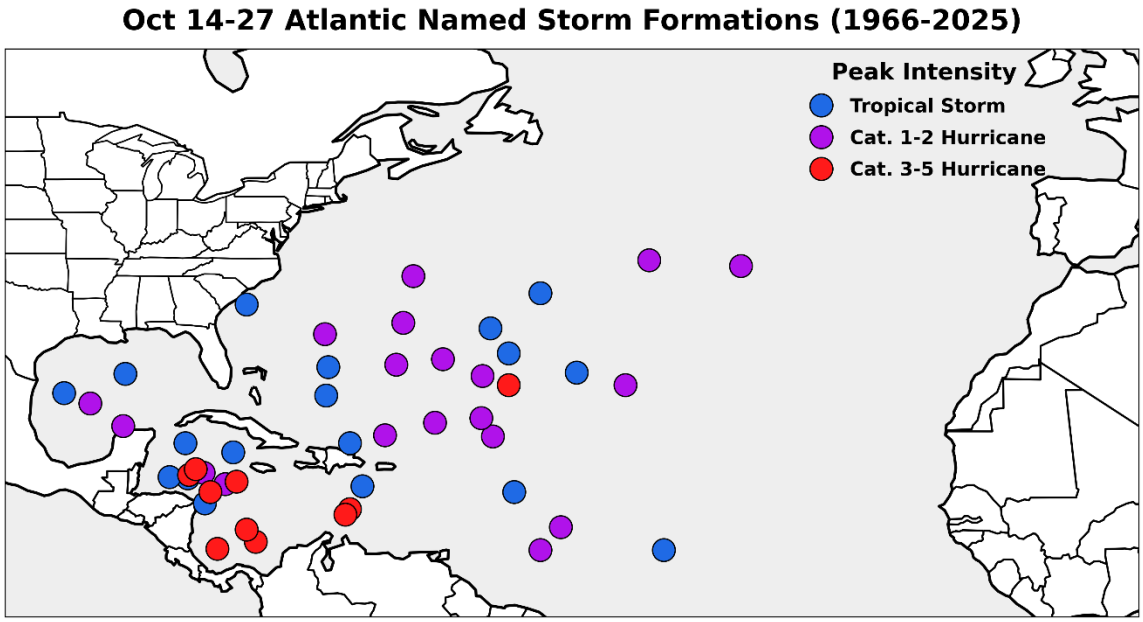


Figure 1: Atlantic named storm formations from October 14–27 from 1966–2025 and the maximum intensity that these named storms reached.

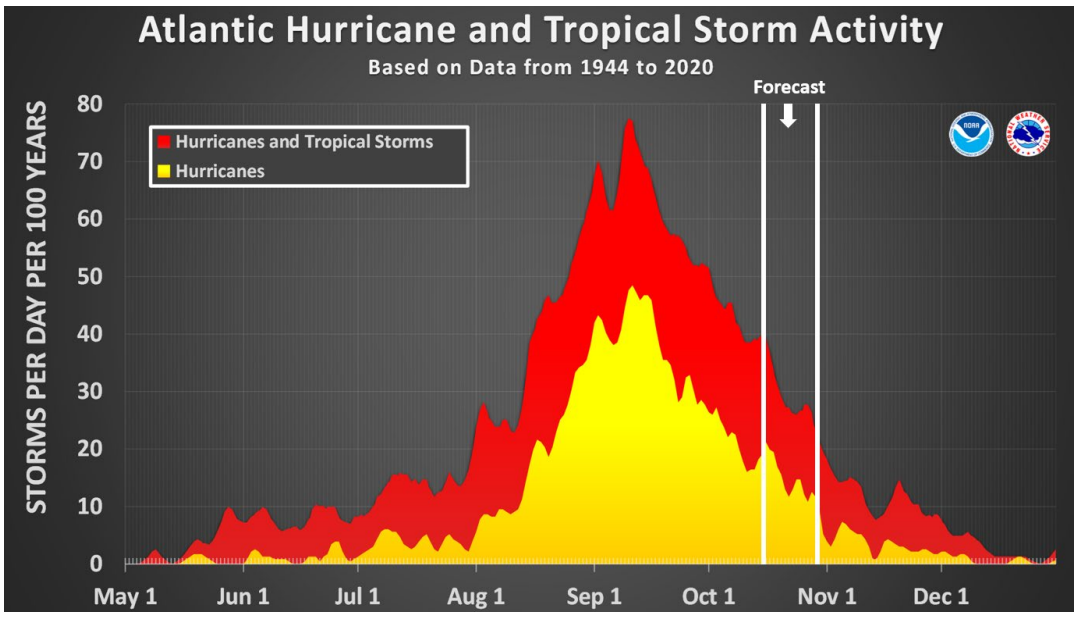


Figure 2: The current forecast period (October 14–27) with respect to climatology, delimited with white lines. Figure courtesy of NOAA.

We now examine how we expect each of the five factors discussed in the introduction will impact Atlantic TC activity for the period from October 14–27.

1) Current Storm Activity

There are no active Atlantic TCs.

2) National Hurricane Center Tropical Weather Outlook

The latest outlook from the National Hurricane Center is not monitoring any areas for TC formation in the next seven days.

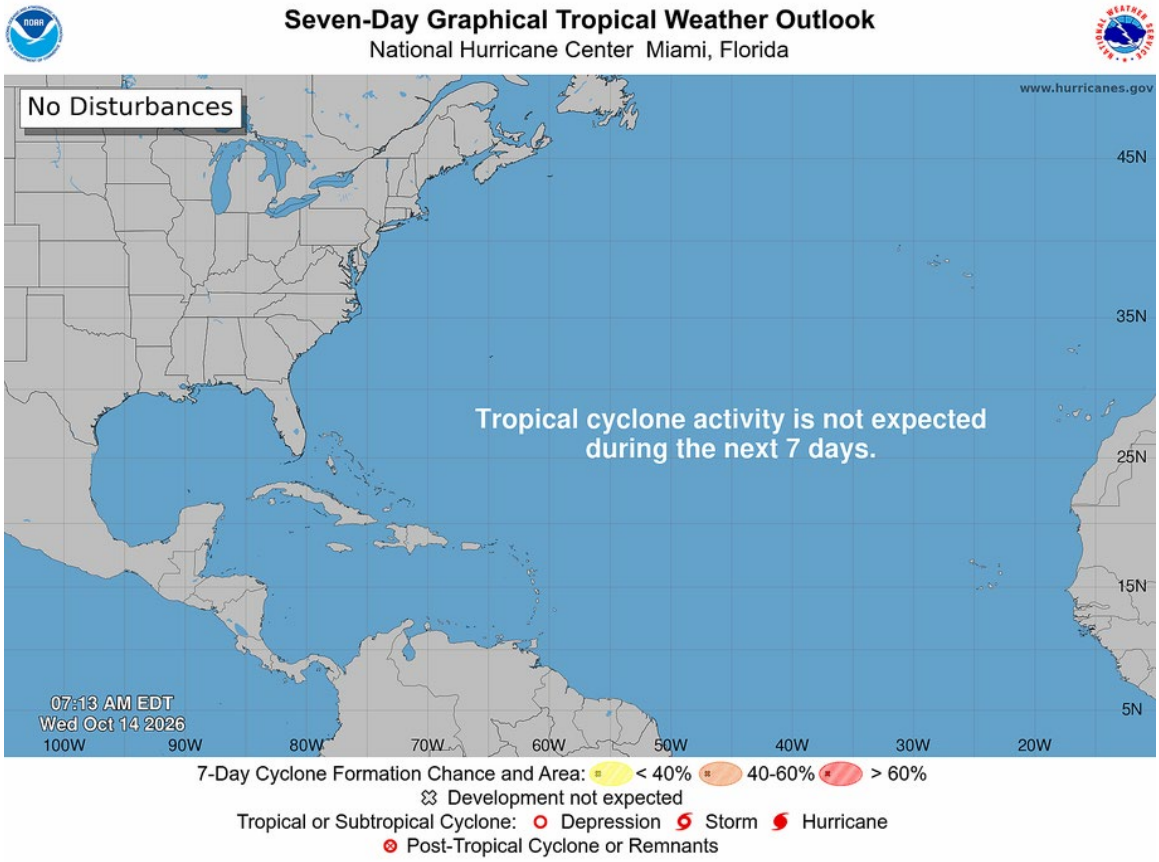


Figure 3: Current National Hurricane Center Atlantic Tropical Weather Outlook.

3) Global Model Analysis

We examine both physics-based dynamical models (ECMWF EPS, GEFS) as well as AI-machine-learning based models (ECMWF AI, Google DeepMind FNV3). Both types of models produce an ensemble of simulations for the next two weeks that we can examine for TC development. There are weak signals for potential TC development in the subtropical Atlantic towards the end of October. The ECMWF EPS ensemble (Figure 4), ECMWF AI ensemble (Figure 5), GEFS ensemble (Figure 6) and Google DeepMind ensemble (Figure 7) show some potential associated with this system, but the overall strength and consistency of the ensemble members are poor.

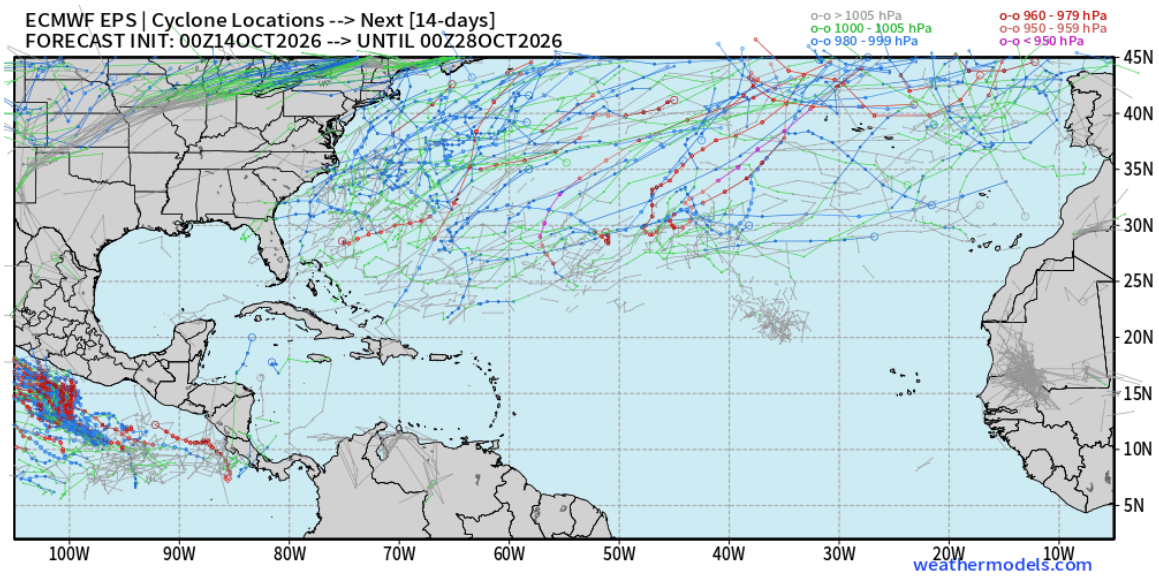


Figure 4: Cyclone locations from the ECMWF EPS ensemble for the next 14 days. Figure courtesy of weathermodels.com.

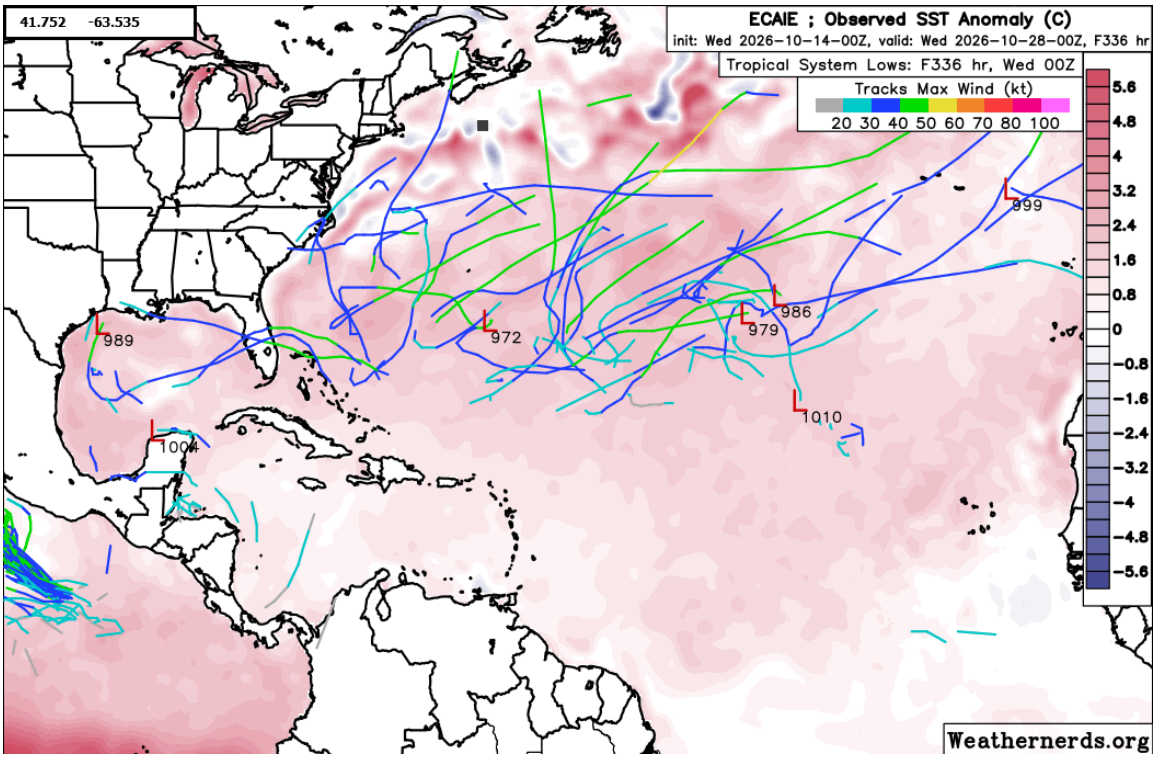


Figure 5: Cyclone locations from the ECMWF AI ensemble for the next 14 days. Figure courtesy of weathernerds.org.

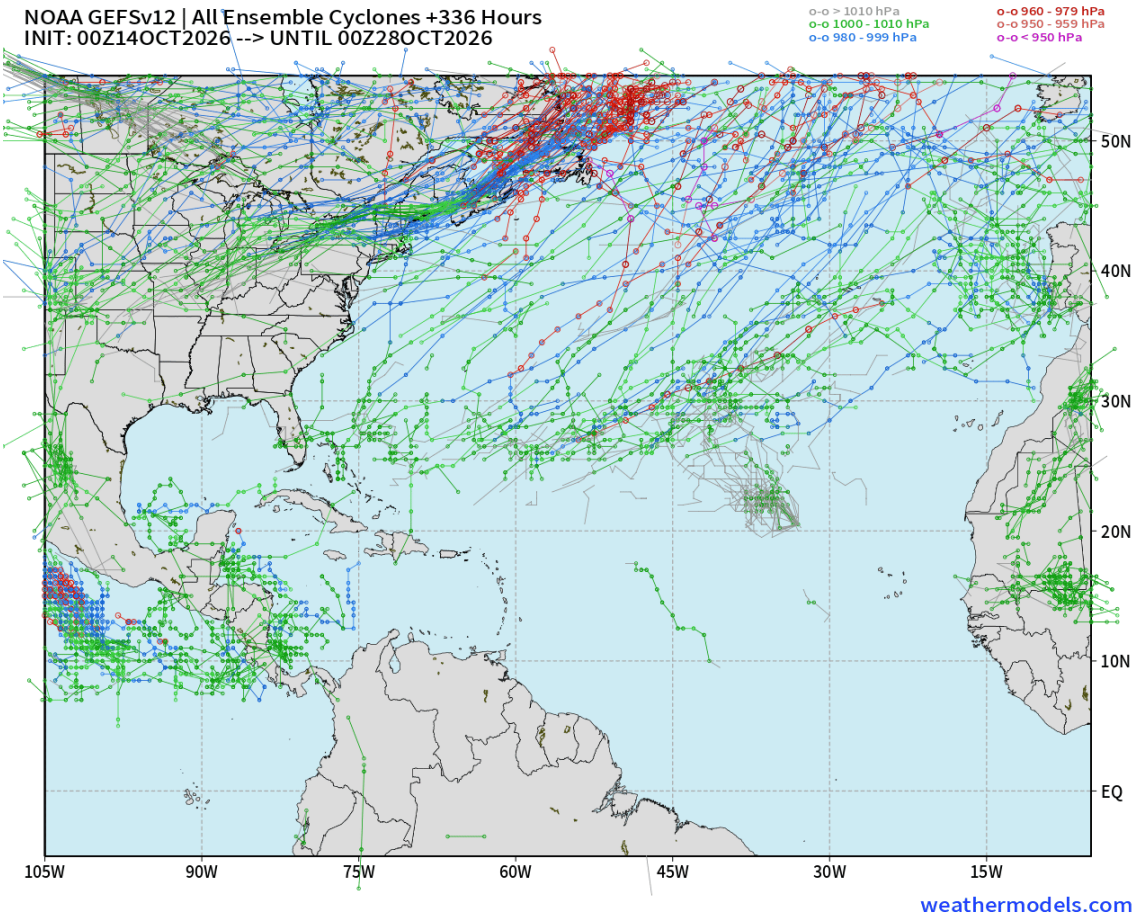


Figure 6: Cyclone locations from the GEFS ensemble for the next 14 days. Figure courtesy of weathermodels.com.

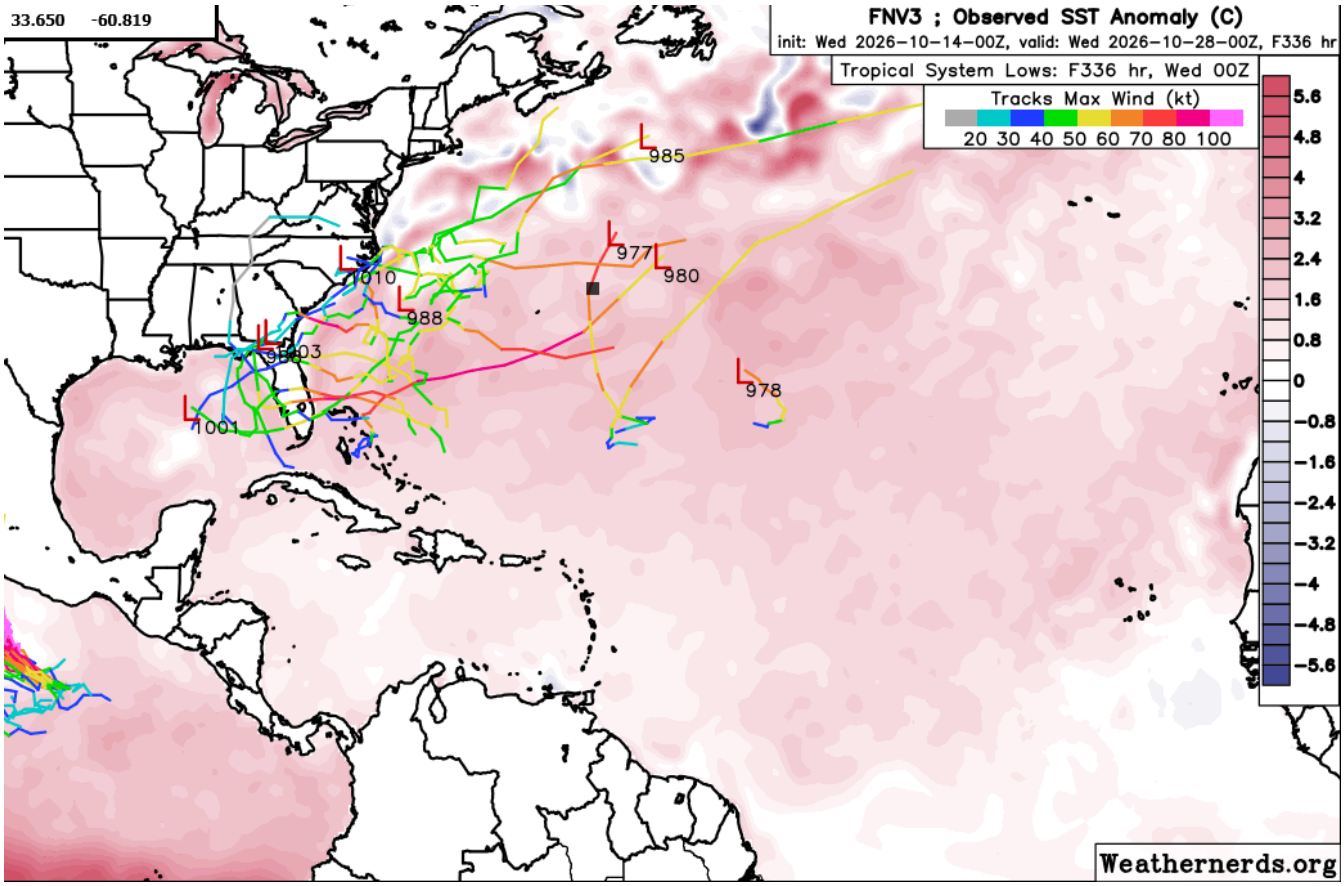


Figure 7: Cyclone locations from the Google DeepMind ensemble for the next 14 days. Figure courtesy of weathernerds.org.

4) Madden-Julian Oscillation

The MJO, as measured by the Wheeler-Hendon index, is currently located over Africa. The EPS generally forecasts the MJO to propagate eastward across Africa and the Indian Ocean during the next couple of weeks (Figure 8). When the MJO is enhancing convection over Africa and the Indian Ocean, Atlantic TC activity is typically elevated.

Despite the somewhat more TC-favorable subseasonal conditions driven by more convectively-favorable phases of the MJO, the strong El Niño is likely to continue to drive above-normal vertical wind shear across the central and western tropical Atlantic, likely still resulting in below-normal activity over the next two weeks (Figure 9).

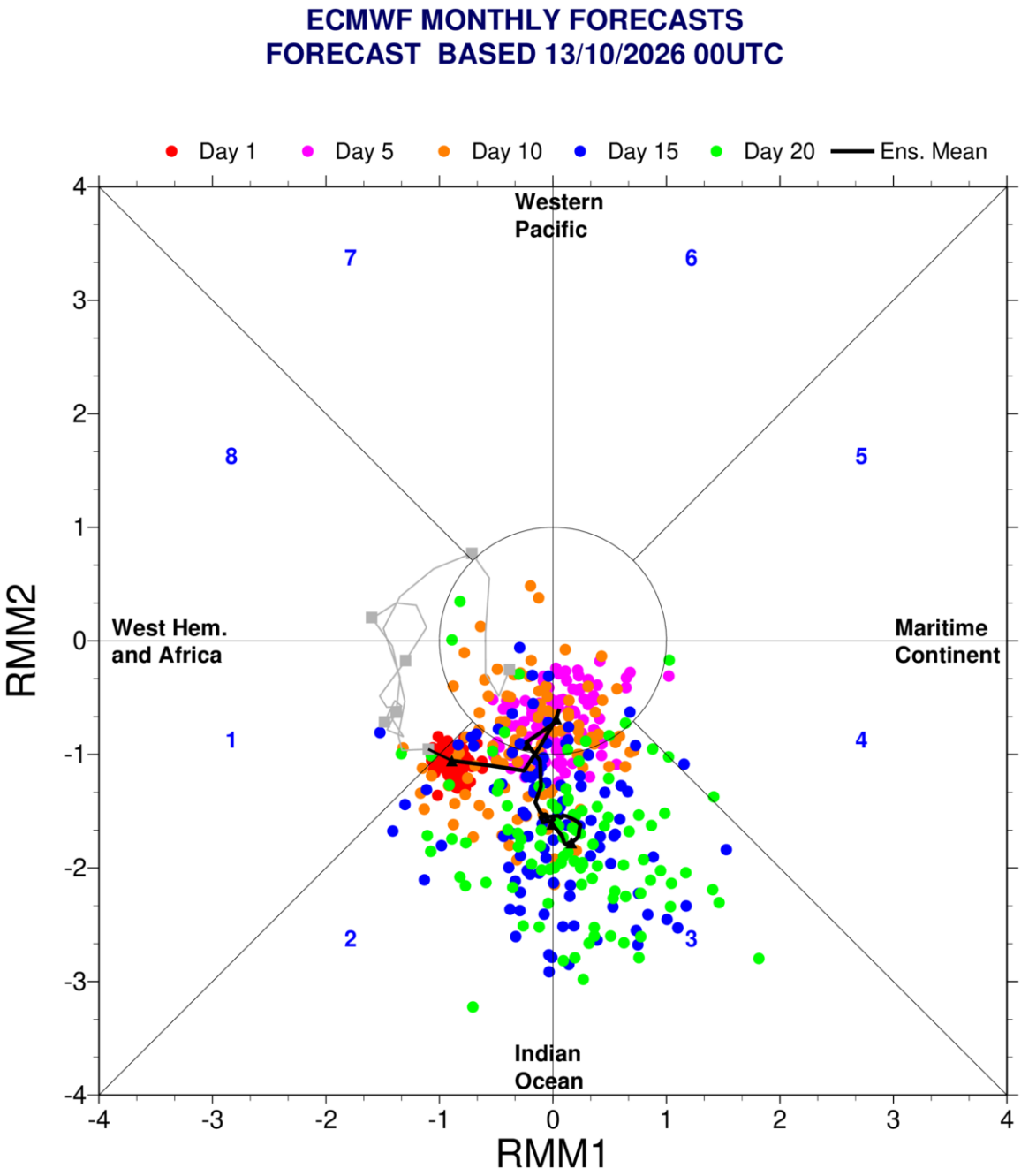


Figure 8: Predicted propagation of the MJO by the EPS. Orange dots correspond to day 10, while blue dots correspond to day 15. Figure courtesy of ECMWF.

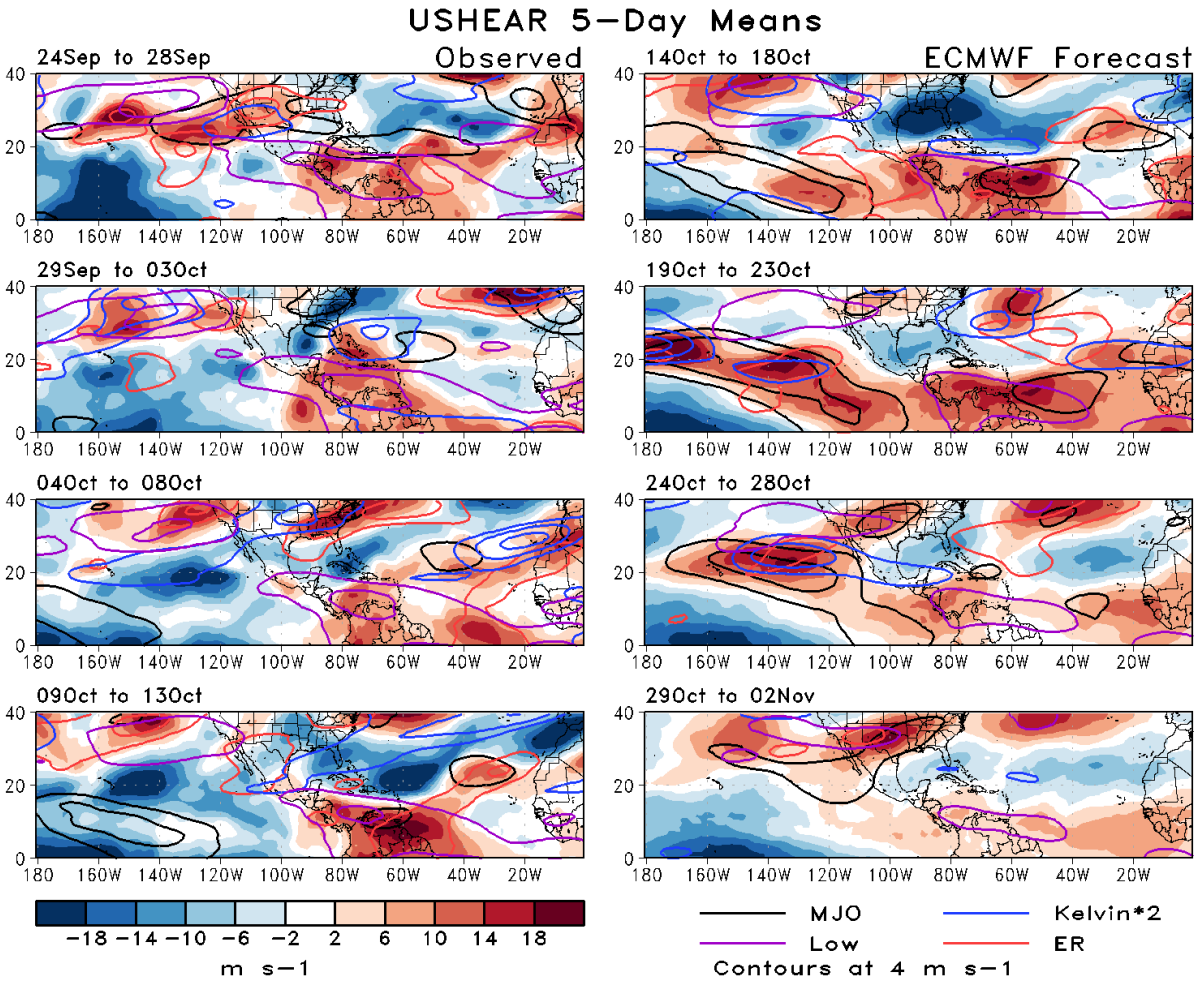


Figure 9: Observed and predicted zonal wind shear by the ECMWF ensemble for the next 20 days. Vertical wind shear is forecast to be above normal (e.g., westerly anomalies) across the tropical central and western Atlantic (where most strong TCs form in mid- to late October). Figure courtesy of Nick Novella (NOAA/Climate Prediction Center).

5) Seasonal Forecast

The 2026 Atlantic hurricane season has been extremely quiet, although we did have an impactful hurricane landfall (Isaias) on 9 October. While subseasonal variability driven by the MJO should make conditions somewhat more conducive relative to climatology than they have been, we expect that the strong El Niño will drive large-scale conditions that should still favor below-normal Atlantic TC activity during the two-week period.

3 Upcoming Forecasts

This is the final two-week forecast of the 2026 Atlantic hurricane season.

VERIFICATION OF SEPTEMBER 30–OCTOBER 13 FORECAST

Normal TC activity occurred during the two-week period from September 30–October 13, with Isaias generating 7 ACE. We had assigned a 78% probability of below-normal activity, with a 20% chance of normal and a 2% chance of above-normal activity during the two-week period.

Table 2 displays the percentage chance that we gave for each category being reached and observed ACE.

Table 2: ACE forecast for TC activity for September 30–October 13, with the probability assigned for each category being reached and the observed ACE.

ACE Category	Definition	Probability in each Category	Observed ACE
Above Normal	Upper Tercile (>12 ACE)	2%	
Normal	Middle Tercile (4–12 ACE)	20%	7
Below Normal	Lower Tercile (<4 ACE)	78%	